

Gamma-knife-based stereotactic radiosurgery for medium- and large-sized posterior uveal melanoma

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Abstract

Background This study investigated the use of gamma-knife-based stereotactic radiosurgery (GKRS) for medium- and large-sized posterior uveal melanoma treatment.

Methods We assessed 50 eyes from 50 consecutive patients with uveal melanoma who were treated with GKRS. All tumors met the criteria for medium- and large-sized uveal melanomas. Patients underwent a single session treatment under local anesthesia; the prescribed radiation dose at the tumor periphery was standardized to 30 Gy at the 50 % isodose field for all patients. The main outcomes were local tumor control, eye retention, and survival rates.

Results The median follow-up time was 40 months (16–78 months). The baseline mean tumor diameter was 10.3 mm (7.1–15.7 mm) and the apical tumor height was 8.7 mm (4.1–16.8 mm). After treatment, the mean tumor diameter was 8.7 mm (5.5–12.0 mm) and the tumor height was 6.2 mm (0.5–11.2 mm). Changes in both tumor height and diameter were statistically significant ($p<0.001$). The tumor control rate was 90 %, and the eye retention rate was 82 %. A total of nine patients (18 %) developed metastasis, and seven (14 %) died due to metastasis during follow-up. Cataracts (34 %) and radiation maculopathy (30 %) were the most frequent complications, and 14 % of patients developed neovascular glaucoma. Visual acuity (VA) decreased significantly after treatment ($p<0.0001$).

Conclusion Treatment using low doses of GKRS, (30 Gy) is an eye-sparing outpatient option for patients with

medium- or large-sized posterior uveal melanomas who are not eligible for brachytherapy or particle therapy. Complications, particularly impaired VA, should be taken into consideration.

Keywords Eye · Tumor · Uvea · Melanoma · Gamma knife · Stereotactic

Introduction

Uveal melanoma (UM) is the most common primary intraocular malignancy in adults. Enucleation has been the traditional treatment for UM. However, over the past few decades, management of this malignancy has shifted toward more conservative eye methods including brachytherapy, charged-particle radiation, local resection, transpupillary thermotherapy, and stereotactic radiotherapy [1–28]. The Collaborative Ocular Melanoma Study (COMS) trial of brachytherapy versus enucleation found no clinically or statistically meaningful differences in mortality rates between groups up to 12 years after treatment [9]. These results encouraged ocular oncologists to treat choroidal melanomas with eye-preserving methods. The advantages of eye-sparing modalities include preserving vision and the cosmetic advantage of avoiding enucleation.

Gamma-knife radiosurgery (GKRS) has been used successfully to treat intracranial neoplasms and vascular lesions [10]. Rand et al. used GKRS to treat melanoma of the eye in animals [11]. In 1989, Zambrano et al. was the first to report results for human patients with UM [12]. Subsequently, several retrospective series investigated GKRS for UM in humans [4–8]. However, few long-term results have been reported. The present study evaluated the outcomes of patients treated with GKRS for medium or large UM.

The manuscript, or parts of it, have not been and will not be submitted elsewhere for publication.

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Material and methods

Subjects

We conducted a retrospective review of the records of 50 consecutive patients with UM treated with GKRS from March 2002 to February 2010 at the Ophthalmology Department of Cerrahpasa Medical School at Istanbul University and the Neurosurgery and Radiation Oncology Departments of the Medical School of Marmara University and Acibadem University. The Institutional Review Board approved the study. Patients with apical tumor heights >3.0 mm were included in the study. Patients with extra ocular involvement or metastasis at the initial examination were excluded.

All patients were examined at the ocular-oncology unit of our clinic and underwent a full ophthalmologic evaluation before treatment, including best-corrected visual acuity (BCVA) measurement, slit-lamp biomicroscopy, intraocular pressure measurement, indirect ophthalmoscopy, color fundus photography, and A- and B-scan ultrasonography (OTI-Scan 1000 ultrasound systems; Ophthalmic Technologies Inc., Toronto, ON, Canada). After treatment, follow-ups were conducted at 1, 3, 6, 9, and 12 months, and then at intervals of 3–6 months using a multidisciplinary approach involving ophthalmologists, neurosurgeons, and radiation oncologists. At each follow-up visit, patients underwent a full ophthalmologic examination, color fundus photography, and A- and B-scan ultrasonography. Metastasis screening included chest radiography, liver function tests, and liver ultrasonography at 6-month intervals. According to the principles of the Declaration of Helsinki, written informed consent was obtained from each study subject. Patients were also informed of all current treatment methods for UM.

Technique

The gamma-knife procedure for UM is well-described in the literature [4–8]. We used the Gamma Unit B model of the Leksell stereotactic system (Elekta Instruments, Stockholm, Sweden). Gamma irradiation was emitted from 201 cobalt

(cobalt-60) sources within a hemisphere around the patient's head, so that each gamma radiation beam converged on a focal unit. We used high-resolution magnetic resonance imaging (MRI) with a voxel size of 1.5 mm³. The precision of combined (imaging and treatment) treatment was ± 0.5 mm.

The patient was positioned prone or supine depending on the tumor location. Patients were administered retrobulbar anesthesia. A Leksell stereotactic head frame was attached to the outer tabula of the skull under local anesthesia. Gamma Plan treatment planning software (Elekta Instruments) was used for MRI-based dose planning (Fig. 1). The maximum dose was delivered to the tumor at the 100 % isodose line. The prescribed radiation dose at the tumor periphery was standardized to 30 Gy at the 50 % isodose field for all patients. Treatment was performed in a single session. After treatment, we performed an MRI to check for eye shifting during the procedure.

Statistical analysis

Statistical analysis was performed by a professional specialist using the NCSS 2007 program. A descriptive analysis was used for tumor characteristics and demographic data. A one-way variance analysis was used to compare the groups, and the χ^2 test was applied to compare qualitative data. An independent *t*-test was used to compare the two groups. The Kaplan–Meier test was applied to assess survival and metastasis. *P*-values less than 0.05 were deemed to indicate statistical significance.

Results

The median age was 53 years (range, 26–78 years) and the median follow-up time was 40 months (range, 16–78 months). Of the 50 tumours, 30 (60 %) were located in right eyes and 20 (40 %) in left eyes; 12 (24 %) were located in the ciliary body, ten (20 %) in the macula, and 18 (36 %) in a mid peripheral region and ten (20 %) in peripapillary area. Table 1 lists demographic data and tumor characteristics, treatment

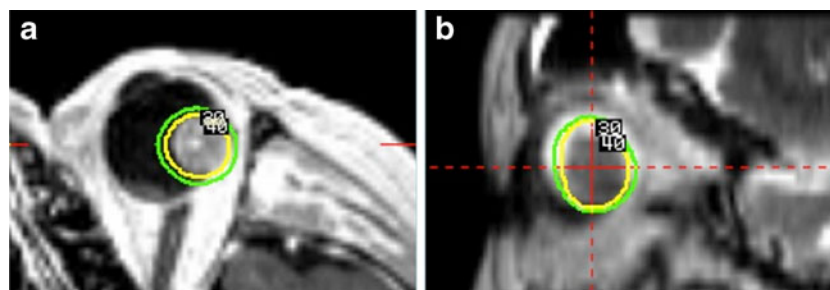


Fig. 1 Stereotactic magnetic resonance imaging of gamma-knife-based stereotactic radiosurgery (GKRS) showing the treatment plan for a tumor 9.98 mm in height and 13.25 mm in basal diameter (T3a

according to AJCC classification). **a** T1 axial image **b** T2 sagittal image. Isodose views were set such that the tumor was entirely encircled by 50 % isodoses (green circles)

Table 1 Demographic data, tumor location and classification based on AJCC with treatment outcomes for each consecutive patient

Patient no	Age	Gender	Involved eye	Follow-up (months)	Tumor location	AJCC classification	Pre TH	Post TH	Treatment response #	Metastasis	Complication	Enucleation
1	63	F	R	51	CB	T 3a	12.0	9.1	1	no	C	0
2	48	F	L	23	Macula	T 1	5.0	3.5	1	no	RM	0
3	60	M	R	42	Macula	T 2a	6.0	2.6	1	no	RON	0
4	53	F	L	46	Mid peripheral	T 3a	16.8	5.5	1	yes	NVG,ERD	1N
5	56	M	R	62	Mid peripheral	T 2b	9.1	7.4	1	no	C	0
6	26	M	L	38	Mid peripheral	T 2b	10.0	8.2	1	no	RR,VH	0
7	78	M	R	52	CB	T 3a	13.1	10.0	1	no	C,RON,ERD	0
8	58	F	R	60	Mid peripheral	T 2a	8.2	7.1	1	no	NVG	0
9	46	M	L	38	Macula	T 1	4.0	2.0	1	no	RM	0
10	55	F	R	52	Mid peripheral	T 2a	8.7	6.2	1	no	C	0
11	55	F	R	62	Macula	T 2a	7.6	4.2	1	yes	RM	0
12	52	M	L	38	CB	T 3a	15.0	5.3	1	no	C,NVG,ERD	1N
13	46	F	L	44	Mid peripheral	T 2b	11.0	9.6	1	no	C	0
14	41	M	R	32	Peripapillary	T 2b	13.0	11.2	1	no	C,NVG,ERD	1N
15	42	F	L	29	Mid peripheral	T 2a	7.4	4.0	1	no	RM	0
16	56	F	R	66	Mid peripheral	T 2a	8.1	5.2	1	no	C	0
17	47	F	R	42	Mid peripheral	T 2a	8.6	5.5	1	no	C	0
18	70	M	R	63	CB	T 3a	9.1	6.3	1	yes	RR,VH	0
19	62	M	L	35	Mid peripheral	T 2a	8.5	7.8	1	no	C	0
20	64	F	R	40	Peripapillary	T 2a	6.2	7.3	2	no	RR	1P
21	58	M	L	56	Peripapillary	T 1	4.1	0.5	1	no	RM,RON	0
22	65	F	L	41	Macula	T 2a	6.2	2.0	1	no	NVG	0
23	60	M	L	56	CB	T 2b	10.6	3.6	1	yes	C	0
24	68	F	L	48	Mid peripheral	T 1	5.7	3.5	1	no	RR	0
25	57	F	R	46	Peripapillary	T 1	5.4	1.9	1	no	RM,RON	0
26	41	M	R	57	Peripapillary	T 2b	9.5	7.1	1	no	RR,RM	0
27	42	F	L	38	Mid peripheral	T 2b	11.0	8.6	1	yes	RR,RM	0
28	67	M	R	36	Macula	T 2a	8.5	6.4	1	no	RM	0
29	72	F	R	68	Mid peripheral	T 2a	8.5	5.3	1	yes	NVG	0
30	37	F	L	44	Macula	T 2a	8.5	9.5	2	no	ERD	1P
31	68	F	L	69	CB	T 2b	9.2	6.0	1	no	NVG	1N
32	52	M	R	30	Peripapillary	T 2b	13.0	4.2	1	no	RON	0
33	27	M	L	22	Macula	T 2a	7.2	6.0	1	no	RM	0
34	59	M	R	70	Peripapillary	T 1	5.4	3.8	1	no	RM,RON	0
35	68	F	R	63	Mid peripheral	T 2b	10.0	8.2	1	yes	RR,VH	0

Table 1 (continued)

Patient no	Age	Gender	Involved eye	Follow-up (months)	Tumor location	AJCC classification	Pre TH	Post TH	Treatment response #	Metastasis	Complication	Enucleation
36	52	M	L	22	CB	T 3a	8.3	5.0	1	no	RR,VH	0
37	43	M	R	26	Mid peripheral	T 2b	10.0	3.0	1	no	RR	0
38	43	M	R	23	Mid peripheral	T 2b	8.0	7.0	1	no	none	0
39	47	F	R	38	CB	T 2b	5.0	6.5	2	no	C	1
40	32	M	R	16	Peripapillary	T 1	4.5	2.5	1	no	C,RR	0
41	72	M	R	23	CB	T 2b	5.0	2.5	1	no	none	0
42	34	M	L	16	Peripapillary	T 2b	11.0	10.0	1	no	ERD	0
43	38	M	R	58	CB	T 3a	9.2	10.7	2	yes	RR	1P
44	26	F	R	36	Mid peripheral	T 2b	9.5	11.0	2	no	C	1P
45	36	M	L	18	Macula	T 2b	9.8	6.7	1	no	RON	0
46	64	M	R	16	Peripapillary	T 2b	10.8	9.8	1	no	RR,VH,ERD	0
47	56	M	R	16	Macula	T 2a	7.4	4.6	1	no	RM	0
48	42	M	R	18	Mid peripheral	T 1	5.5	5.0	1	no	none	0
49	68	M	R	16	CB	T 2b	8.2	7.6	1	no	ERD	0
50	66	M	L	16	CB	T 3a	11.2	9.8	1	yes	C	0

AJCC: # American Joint Commission on Cancer * Classification was made based on AJCC Cancer Staging Manual. 7th ed. Springer, New York, NY, 2009

M: male F: female R: right L: left

Pre TH: tumor apical height before treatment

Post TH: tumor apical height after treatment

CB: tumor epicenter was in ciliary body

Macula: tumor located in the macula and posterior margin was far from 2 optic disc diameter from optic disc

Peripapillary: tumor posterior margin was within 2 disc diameter of optic disc

Mid Peripheral: tumor posterior margin was between 2 disc diameter from optic disc and equator and not located in the macula

1 = same size or smaller after treatment; 2 = tumor continued to increase in size

C: cataract RM: radiation maculopathy RON: radiation optic neuropathy RR: radiation retinopathy VH: vitreous hemorrhage

ERD: exudative retinal detachment NVG: neovascular glaucoma

IN: enucleation due to neovascular glaucoma

1P: enucleation due to progression

Table 2 Summarized data of the main outcomes of the study

	Before treatment	After treatment
Tumor thickness (mm) (mean,range)	8.7 (4.1–16.8)	6.2 (0.5–11.2)
Tumor diameter (mm) (mean,range)	10.3 (7.1–15.8)	8.7 (5.5–12.0)
Visual acuity (mean)	20/60	20/400
	<i>N</i>	%
Treatment response		
Yes	45	90
No	5	10
Metastasis		
Liver	6	12
Liver and bone	2	4
Multiple	1	2
Enucleation		
Due to progression	5	10
Due to NVG	4	8
Death ^{##}	7	14

^{##} number of patients who were dead at the end of follow-up period

outcomes and complications. Of the tumours, 8 (16 %), 15 (30 %), 19 (38 %), and 8 (16 %) tumours were classified as T1, T IIa, T IIb, or T IIIa respectively (based on the American Joint Committee on Cancer [AJCC] Cancer Staging Manual, Seventh Edition) (Table 1). The changes in apical height and tumor diameter following GKRS were statistically significant (Table 2) ($p<0.001$). Treatment response was classified as ‘good’ (same size or smaller after treatment) (Fig. 2) or ‘failure’ (tumor continued to increase in size). No significant relationships were observed between age, gender, laterality, tumor location, tumor apical height, maximum tumor base diameter, American Joint Commission on Cancer (AJCC) classification, or treatment type ($p>0.1$ for all). Although tumors responded well, four patients developed uncontrolled neovascular glaucoma (NVG) and also underwent enucleation. Local tumor control was achieved in 90 % of patients

during the follow-up period. Eye retention rates were 88 % and 82 % for 3 and 5 years respectively. The 3- and 5-year overall survival rates were 94 % and 86 % respectively. Of the 50 patients, nine (18 %) developed distant metastasis, and seven patients died due to metastasis during follow-up. The metastasis-free survival rate for 1, 2, and 5 years was 98 %, 93 %, and 65 % respectively. The median time for metastasis-free survival was 63.75 months (Fig. 3). No statistical relationship was observed between age, gender, laterality, tumor apical height, tumor location, maximum tumor base diameter, AJCC classification, distant metastasis, or 3- and 5 year survival rates ($p>0.1$ for all).

Complications and visual acuity (VA)

Table 1 described the complications for each patient. The most frequent patient complaint was pain because of head frame fixation. A significant relationship was observed between complications and tumor location ($p=0.001$ for all). Cataract development occurred more frequently with anterior tumors. Radiation retinopathy, maculopathy, and optic neuropathy were also more prominent in more posterior-located tumors. Acute retinal exudation (ARE) was more frequent in patients with high-volume tumors, and was seen mostly within 2 years. If ARE involved the macula, it caused decreased or poor vision. Most cases resolved by themselves, however in three patients it didn’t resolve. Those patients had also uncontrolled NVG and underwent enucleation. NVG was the worst complication, and caused a painful eye that was generally managed with enucleation, occurred most often within the first 3 years, and was associated with high-volume and anteriorly-located tumors. A significant absolute reduction in visual acuity was observed during follow-up ($p<0.0001$). At the time of diagnosis, the median BCVA in eyes with UM was 20/60, with 40 % of eyes at 20/50 or better and 28 % of eyes at 20/200 or worse visual acuity. After GKRS, the median BCVA was 20/400, with 15 % of patients at 20/80 or better and 60 % at 20/200 or worse visual acuity (Table 3).

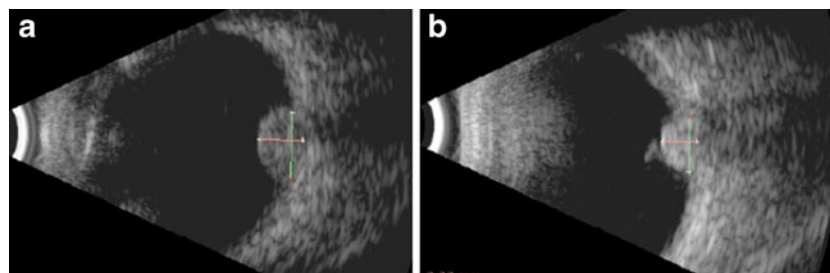


Fig. 2 Pre- and post-treatment B-scan ultrasound images of a patient treated with gamma-knife-based stereotactic radiosurgery (GKRS). Before GKRS, the mass was acoustically hollow by B-scan

ultrasonography (a). After treatment, both the height and basal diameter decreased, and the mass became acoustically solid on B-scan ultrasonography (c)

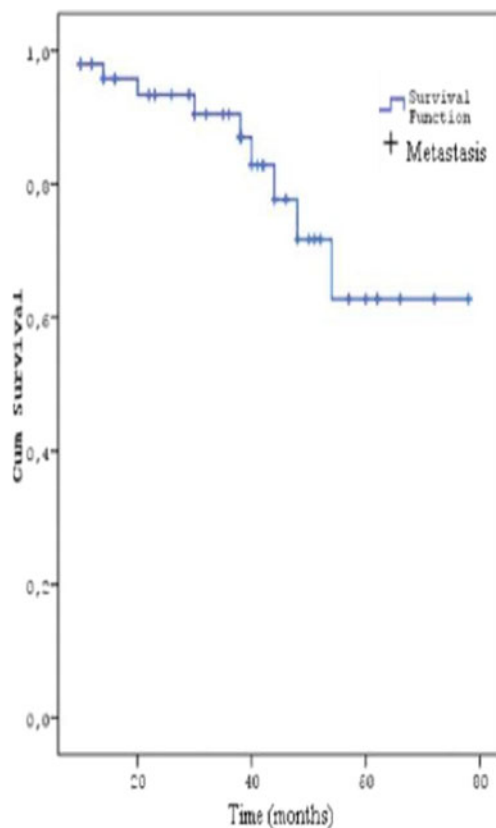


Fig. 3 Kaplan–Meier cumulative survival plot for metastasis free survival (MFS) in patients treated with gamma-knife-based stereotactic radiosurgery (GKRS) for posterior uveal melanoma

Discussion

Several investigators have reported favorable results with GKRS for UM. Previous studies have used widely varying doses of GKRS to treat UM in a heterogeneous patient population. Earlier series used higher radiation doses, but doses were gradually de-escalated to encourage lower complication rates [4–8]. A few studies have reported long-term follow-up results for medium- and large-sized UMs using low-dose radiation. Dinca et al. reported long-term follow-up results regarding dose de-escalation. They used different peripheral doses ranging from 70 to 35 Gy [6]. Although they reported that 35 Gy yielded lower complication rates, they did not mention tumor sizes separately for 35 Gy. Modorati et al. reported a survival rate of 88.8 % at 3 years and 81.9 % at 5 years for patients with medium-sized UMs [4]. They also used various radiation doses between 50 and 35 Gy, and reported different survival rates for different doses during the follow-up period. Mueller et al. used GKRS to treat patients with large UMs [5]. They applied a uniform low-dose radiation (a 50 % isodose set at 25 Gy) and achieved tumor control in 97 % of eyes and a 94 % eye retention rate. However, their follow-up time was rather

short (the median duration was 12 months and the longest was 20 months among 35 patients). Our study was unique in that it used a low-dose standard peripheral dose—30 Gy at the 50 % isodose line for all patients—with a 40-month median follow-up period for medium- and large-sized melanomas.

In the COMS trial, the 5-year death rate with a histopathologically confirmed melanoma metastasis was 10 % in the I-125 brachytherapy arm and 11 % in the enucleation arm [9]. Studies have demonstrated that other well-established treatment modalities such as I-125 plaque brachytherapy and proton beam radiotherapy (PBRT) provide good outcomes [13–17]. In our series metastasis, 5-year tumor control, and eye preservation rates were 14 %, 90 %, and 82 % respectively. We found the local tumor control and metastasis rates were in the range of previous findings of treatment with GKRS (Table 4) and other treatment modalities (Table 5) [4–8]. The main cause for the lower eye preservation rate in the present study was the uncontrolled NVG with painful eyes that required enucleation.

We found no statistical difference between the survival rate and patient or tumor features such as AJCC classification, age, tumor size (diameter and height), and tumor location. This was probably due to the relatively low number of cases among subgroups.

The radiation dose was delivered in a single fraction. Because our treatment was applied in a single session, patients tolerated it very well, and none exhibited a significant shifting of the globe during the control MRI compared with the MRI obtained before treatment. GKRS has an advantage over proton beam and helium ion radiotherapy in that it spares the patient two surgical procedures for insertion and removal of the tantalum rings used for localization of the tumor margin. Additionally, during eye plaque therapy, one or more muscles may be disinserted during plaque placement, and require a second surgery for eye plaque removal [23]. GKRS also prevents the need for 3–4 days of hospitalization, which is required for radiation protection during plaque brachytherapy. Thus, GKRS is a less invasive approach.

Another important aspect of GKRS concerns ocular complications. Various complication rates have been reported in series with GKRS and other different treatment modalities; our complication rates were comparable with previous findings (Tables 4 and 6) [4–8, 18–22, 23–26]. In our study, the most frequent complication was cataracts. This could be the result of tumor localization, because 60 % of tumors were more anteriorly located. The most disturbing complication was NVG. Some cases could be controlled with medication or surgery, but four patients underwent enucleation because of a painful eye, and NVG accounted for 45 % of enucleation. Radiation retinopathy and optic neuropathy were

Table 3 Visual acuity of patients before and after treatment with main reason for decreased or poor vision for each patient

Patient no	Age (years)	Follow-up (months)	Pre-treatment vision	Vision at last visit	Main reason for decreased or poor vision
1	63	51	20/40	20/2,000	C
2	48	23	20/80	20/400	RM
3	60	42	20/80	20/400	RON
4	53	46	20/20,000	enuc	Enuc due to NVG
5	56	62	20/2,000	20/2,000	C
6	26	38	20/40	20/60	VH
7	78	52	20/40	20/2,000	C,RON
8	58	60	20/200	20/2,000	NVG
9	46	38	20/40	20/100	RM
10	55	52	20/50	20/60	C
11	55	62	20/125	20/2,000	RM
12	52	38	20/400	enuc	Enuc due to NVG
13	46	44	20/50	20/80	C
14	41	32	20/2,000	enuc	Enuc due to NVG
15	42	29	20/63	20/80	RM
16	56	66	20/40	20/2,000	C
17	47	42	20/32	20/50	C
18	70	63	20/63	20/2,000	VH
19	62	35	20/50	20/200	C
20	64	40	20/63	enuc	Enuc due to progression
21	58	56	20/63	20/400	RM
22	65	41	20/50	20/200	NVG
23	60	56	20/80	20/400	C
24	68	48	20/25	20/32	SMD
25	57	46	20/80	20/2,000	RM,RON
26	41	57	20/200	20/2,000	RM
27	42	38	20/100	20/2,000	RM
28	67	36	20/20,000	20/20,000	RM
29	72	68	20/63	20/2,000	NVG
30	37	44	20/2,000	enuc	Enuc due to progression
31	68	69	20/63	enuc	Enuc due to NVG
32	52	30	20/2,000	20/2,000	RON
33	27	22	20/100	20/400	RM
34	59	70	20/40	20/200	RON
35	68	63	20/63	20/400	VH
36	52	22	20/32	20/100	VH
37	43	26	20/50	20/2,000	RR
38	43	23	20/100	20/100	Stable
39	47	38	20/63	enuc	Enuc due to progression
40	32	16	20/40	20/60	C
41	72	23	20/25	20/30	SMD
42	34	16	20/400	20/2,000	ERD
43	38	58	20/2,000	enuc	Enuc due to progression
44	26	36	20/100	enuc	Enuc due to progression
45	36	18	20/2,000	20/2,000	RON
46	64	16	20/2,000	20/2,000	VH
47	56	16	20/20,000	20/20,000	RM
48	42	18	20/32	20/32	stable
49	68	16	20/32	20/100	ERD
50	66	16	20/50	20/100	C

Enuc: enucleation *C*: cataract *RM*: radiation maculopathy *RR*: radiation retinopathy

VH: vitreous hemorrhage *ERD*: exudative retinal detachment *SMD*: senile macular degeneration

NVG: neovascular glaucoma *RON*: radiation optic neuropathy

Table 4 Comparison of reported complications of published GKRS studies with current study

	No of patients	Mean radiation dose (Gy)	Tumor thickness (mean, mm)	Follow-up (mean, months)	Tumor control	Metastasis	Radiation cataract	Radiation retinopathy	Radiation papillopathy	Neovascular glaucoma
Current study Gamma knife	50	30 (50 % isodose)	8.7	40 (median)	90 %	18 %	30 %	24 %	14 %	14 %
Packer et al. [18] Iodine 125	65	91	NA	64	92 %	17.20 %	45.3 %	23.40 %	NA	10.90 %
Lumbruso et al. [19] Proton beam	1062	60	5.4	38	97 %	15 %	22 %	47 %	15 %	15 %
Char et al. [20] Helium	164	69.3	6.5	61	93 %	18 %	8 %	46 %	27 %	NA
Finger [21] Pallidium 103	400	73	NA [#]	51	97 %	6 %	NA	22 %	NA	2.50 %
Rouberol et al. [22] Ru 106	213	60	5	100 (median)	84 %	20 %	12 %	17 %	8.50 %	7 %

[#] 92 % of patients were medium-sized due to COMS

Table 5 Comparison of our study outcomes with other treatment modalities

	Patients (n)	Tumor thickness (median, mm)	Median f-u, months (range)	Dose (Gy)	Eye retention (%)	Local control (%)	Metastasis (%)	Survival rates at 5 years
Current study	50	8.7	40 (16–70)	30	82	90	18	86
Mueller et al. [5]	35	9.1	NA (10–20)	25	94	97	NA	NA
Simonova et al. [7]	75	8.5	32 (10–74)	20–76	81	87	21	NA
Modorati et al. [4]	78	6.1	31.3 (NA)	30–50	89.7	91	12	81.9
Fakiris et al. [8]	19	4.1	40 (7–81)	40	94.4	94.4	32	55
Dinca et al. [6]	170	NA	63.5 (6–240)	35–70	84	NA	NA	64.5

Table 6 Comparison of tumor features, treatment dose and outcomes of current study with published GKRS studies

	Radiation dose (Gy)	Tumor thickness (median, mm)	Median f-u, months (range)	Radiation retinopathy (%)	Radiation maculopathy (%)	Radiation papillopathy (%)	NVG (%)	Cataract (%)
Current study	30	8.7	40 (16–70)	24	30	14	14	30
Modorati et al. [4]	30–50	6.1	31.3	13.5	NA	15.5	19	6.2
Mueller et al. [5]	25	9.1	NA (10–20)	2	NA	14	6	6
Simonova et al. [7]	20–76	8.5	32 (10–74)	NA	NA	10	18	NA
Fakiris et al. [8]	40	4.1	40 (7–81)	11	NA	11	0	NA
Dinca et al. [6]	35–70	NA	63.5 (6–240)	26 (35 Gy) 42 (50–70 Gy)	NA NA	16 (35 Gy) 21 (50–70 Gy)	8 (35 Gy) 20 (50–70 Gy)	11 (35 Gy) 20 (50–50 Gy)

NVG: Neovascular glaucoma *f-u*: follow-up

observed more in tumors located in the macula and peripapillary region. Visual acuity (VA) decreased significantly at the end of follow-up ($p < 0.0001$) and BCVA was 20/200 or worse in 60 % of patients (Table 3). In COMS report no. 16, the median VA was 20/125 at 3 years after I-125 brachytherapy, with 45 % having 20/200 or worse visual acuity within 3 years of treatment [28]. In our study, cataracts, NVG, vitreous hemorrhage, exudative retinal detachment, and radiation maculopathy were responsible for most of the vision losses.

In conclusion, although GKRS does not offer better results for UM than brachytherapy or particle therapy, its results for local tumor control as well as eye retention and survival rates are favorable. However, the high complication rates, particularly the significant decrease in vision, highlight the need to use this treatment cautiously. We believe that GKRS is an ocular-conserving option that may be considered for treatment of posterior UM in carefully selected patients, and must be saved only for cases where a patient cannot go to the operating room or access treatment by brachytherapy or PBRT. Randomized controlled prospective studies, which compare GKRS with brachytherapy and/or PBRT, will provide us with additional data.

Conflict of interest The authors have no proprietary interest.

References

- Shields CL, Cater J, Shields JA, Chao A, Krema H, Materin M, Brady LW (2002) Combined plaque radiotherapy and transpupillary thermotherapy for choroidal melanoma: tumor control and treatment complications in 270 consecutive patients. *Arch Ophthalmol* 120:933–940
- Win PH, Robertson DM, Buettner H, McCannel CA, Bennett SR (2006) Extended follow-up of small melanocytic choroidal tumors treated with transpupillary thermotherapy. *Arch Ophthalmol* 124:503–506
- Puusaari I, Damato B, Kivelä T (2007) Transscleral local resection versus iodine brachytherapy for uveal melanomas that are large because of tumor height. *Graefes Arch Clin Exp Ophthalmol* 245:522–533
- Modorati G, Miserocchi E, Galli L, Picozzi P, Rama P (2009) Gamma knife radiosurgery for uveal melanoma: 12 years of experience. *Br J Ophthalmol* 93:40–44
- Mueller AJ, Talies S, Schaller UC, Horstmann G, Wowra B, Kampik A (2000) Stereotactic radiosurgery of large uveal melanomas with the gamma-knife. *Ophthalmology* 107:1381–1387
- Dinca EB, Yianni J, Rowe J, Radatz MW, Preitui-Pietro D, Rundle P, Rennie I, Kemeny AA (2012) Survival and complications following Gamma Knife radiosurgery or enucleation for ocular melanoma: a 20-year experience. *Acta Neurochir (Wien)* 154:605–610
- Simonova G, Novotny J Jr, Liscak R, Pilbauer J (2002) Leksell gamma knife treatment of uveal melanoma. *J Neurosurg* 97:635–639
- Fakiris AJ, Lo SS, Henderson MA, Witt TC, Worth RM, Danis RP, Des Rosiers PM, Timmerman RD (2007) Gamma-knife-based stereotactic radiosurgery for uveal melanoma. *Stereotact Funct Neurosurg* 85:106–112
- Collaborative Ocular Melanoma Study Group (2006) The COMS randomized trial of iodine-125 brachytherapy for choroidal melanoma: V. Twelve-year mortality rates and prognostic factors: COMS report No. 28. *Arch Ophthalmol* 124:1684–1693
- Suh JH (2010) Stereotactic radiosurgery for the management of brain metastases. *N Engl J Med* 362:1119–1127
- Rand RW, Khonsary A, Brown WJ, Winter J, Snow HD (1987) Leksell stereotactic radiosurgery in the treatment of eye melanoma. *Neurol Res* 9:142–146
- Zambrano AD, Chinela AB, Bunge HJ (1989) Stereotactic radiosurgery for uveal melanomas. Protocol for treatment. *Archivos Ophthalmol Buenos Aires* 64:49–54
- Gragoudas E, Li W, Goitein M, Lane AM, Munzenrider JE, Egan KM (2002) Evidence-based estimates of outcome in patients irradiated for intraocular melanoma. *Arch Ophthalmol* 120:1665–1671
- Macdonald EC, Cauchi P, Kemp EG (2011) Proton beam therapy for the treatment of uveal melanoma in Scotland. *Br J Ophthalmol* 95:1691–1695
- Quivey JM, Augsburger J, Snelling L, Brady LW (1996) I-125 plaque therapy for uveal melanoma. Analysis of the impact of time and dose factors on local control. *Cancer* 77:2356–2362
- Shields CL, Shields JA, Karlsson U, Menduke H, Brady LW (1990) Enucleation after plaque radiotherapy for posterior uveal melanoma. Histopathologic findings. *Ophthalmology* 97:1665–1670
- Shields CL, Naseripour M, Cater J, Shields JA, Demirci H, Youseff A, Freire J (2002) Plaque radiotherapy for large posterior uveal melanomas (≥ 8 -mm thick) in 354 consecutive patients. *Ophthalmology* 109:1838–1849
- Packer S, Stoller S, Lesser ML, Mandel FS, Finger PT (1992) Long-term results of iodine-125 irradiation of uveal melanoma. *Ophthalmology* 99:767–773
- Lumbroso L, Levy C, Plancher C, Frau E, D'hermies F, Schlienger P, Dendale R, Mammar H, Delacroix S, Nauraye C, Noel G, Ferrand R, Desblancs C, Mazal A, Validire P, Asselain B, Desjardins L (2002) Results of proton beam irradiation for treatment of choroidal melanoma. *J Fr Ophtalmol* 25:290–297
- Char DH, Castro JR, Kroll SM, Irvine AR, Quivey JM, Stone RD (1990) Five-year follow-up of helium ion therapy for uveal melanoma. *Arch Ophthalmol* 108:209–214
- Finger PT, Chin KJ, Duvall G (2009) Palladium-103 for Choroidal Melanoma Study Group. Palladium-103 ophthalmic plaque radiation therapy for choroidal melanoma: 400 treated patients. *Ophthalmology* 116:790–796
- Rouberol F, Roy P, Kodjikian L, Gerard JP, Jean-Louis B, Grange JD (2004) Survival, anatomic, and functional long-term results in choroidal and ciliary body melanoma after ruthenium brachytherapy (15 years' experience with beta-rays). *Am J Ophthalmol* 137:893–900
- Sener EC, Kiratli H, Gedik S, Sanac AS (2004) Ocular motility disturbances after episcleral plaque brachytherapy for uveal melanoma. *J AAPOS* 8:38–45
- Jones R, Gore E, Mieler W, Murray K, Gillin M, Albano K, Erickson B (2002) Posttreatment visual acuity in patients treated

- with episcleral plaque therapy for choroidal melanomas: dose and dose rate effects. *Int J Radiat Oncol Biol Phys* 52:989–995
25. Jensen AW, Petersen IA, Kline RW, Stafford SL, Schomberg PJ, Robertson DM (2005) Radiation complications and tumor control after I-125 plaque brachytherapy for ocular melanoma. *Int J Radiat Oncol Biol Phys* 63:101–108
26. Fontanesi J, Meyer D, Xu S, Tai D (1993) Treatment of choroidal melanoma with I-125 plaque. *Int J Radiat Oncol Biol Phys* 26: 619–623
27. Gragoudas ES, Li W, Lane AM, Munzenrider J, Egan KM (1999) Risk factors for radiation maculopathy and papillopathy after intraocular irradiation. *Ophthalmology* 106:1571–1577
28. Melia BM, Abramson DH, Albert DM, Boldt HC, Earle JD, Hanson WF, Montague P, Moy CS, Schachat AP, Simpson ER, Straatsma BR, Vine AK, Weingeist TA, Collaborative Ocular Melanoma Study Group (2001) Collaborative ocular melanoma study (COMS) randomized trial of I-125 brachytherapy for medium choroidal melanoma. I. Visual acuity after 3 years COMS report no. 16. *Ophthalmology* 108:348–366

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